

1. Why do we need an activation function in a neural network? What happens if we don't use one?

Answer :

We need an activation function in a neural network to add non-linearity so that the model can learn complex patterns. If we don't use any activation function then the whole neural network becomes just a simple linear model, no matter how many layers we add, it behaves like one straight line.

2. Explain why sigmoid is called a "squashing function".

Answer :

The sigmoid is called a squashing function because it squeezes any input into a small range 0 to 1. The big positive numbers are close to 1 and big negatives are close to 0. Because of its fixed range compressing nature, it is called a squashing function.

3. What is the difference between Sigmoid and Tanh activation functions?

Answer :

The output range of *Sigmoid* is 0 to 1, for *Tanh* the range is -1 to 1. *Sigmoid* is non centered always positive but *Tanh* centered at 0. *Sigmoid* is better for probabilities and *Tanh* is better for the hidden layer. Overall *Tanh* is better because it covers negative + positive both values which helps learning.

4. Why is ReLU more popular than Sigmoid in deep neural networks?

Answer :

Rectified Linear unit (ReLU) is faster, simpler, works better in deep networks, and the sigmoid is very flat which slows down learning. This is why ReLU is more popular than sigmoid in deep neural networks.

5. What problem does ReLU solve compared to Sigmoid?

Answer :

ReLU helps solve the vanishing gradient problem. In sigmoid gradient becomes very small, learning almost stops in deep layers. In ReLU, it keeps gradients strong for positive values, so training is faster and more effective.

6. When should we use Softmax instead of Sigmoid?

Answer :

For yes / no like binary classification we should use sigmoid. For multi-class classification like one correct out of many, we need to use softmax.

7. Why is Binary Cross Entropy preferred over MSE for classification?

Answer :

Binary Cross Entropy (BCE) is designed for classification, it penalizes wrong confident predictions strongly. On the other hand, Mean Squared Error (MSE) treats problems like regression, it is slower learning for classification. BCE is preferred over MSE because of its better gradients and faster learning.

8. If true label $y = 1$ and predicted probability $\hat{y} = 0.9$, calculate the Binary Cross Entropy loss.

Answer :

BCE formula :

$$L = - \left[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}) \right]$$

$$\text{Given, } y = 1, \hat{y} = 0.9$$

$$\text{So, } L = - \log(0.9) \approx 0.105$$

9. If $y = 0$ and $\hat{y} = 0.8$, calculate the loss.

Answer :

We know from BCE , loss :

$$L = - \left[y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}) \right]$$

$$\text{Given, } y = 0, \hat{y} = 0.8$$

$$\text{So, } L = - \log(1 - 0.8) \approx 1.609$$

10. Compute $\text{sigmoid}(0)$, $\text{sigmoid}(2)$, and $\text{sigmoid}(-2)$.

Answer :

We know *Sigmoid* formula :

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\sigma(0) = 0.5$$

$$\sigma(2) \approx 0.88$$

$$\sigma(-2) \approx 0.12$$